
Verbosity Bias in Preference Labeling by Large Language Models

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Abstract

In recent years, Large Language Models (LLMs) have witnessed a remarkable surge in prevalence, altering the landscape of natural language processing and machine learning. One key factor in improving the performance of LLMs is alignment with humans achieved with Reinforcement Learning from Human Feedback (RLHF), as for many LLMs such as GPT-4, Bard, etc. In addition, recent studies are investigating the replacement of human feedback with feedback from other LLMs named Reinforcement Learning from AI Feedback (RLAIF). We examine the biases that come along with evaluating LLMs with other LLMs and take a closer look into verbosity bias – a bias where LLMs sometimes prefer more verbose answers even if they have similar qualities. We see that in our problem setting, GPT-4 prefers longer answers more than humans. We also propose a metric to measure this bias.

1 Introduction

Large Language Models (LLMs) have made tremendous strides in recent years and continue to gain popularity (Zhao et al., 2023). With its growing size in network parameters, its wide application ranges from conventional natural language processing tasks such as chat-bots, summarization, and translation, to other applications beyond its original intended use such as search engines, programming assistance, and foundation models (Zhao et al., 2023; Brants et al., 2007; Katz, 1987).

After pretraining for general purposes, LLMs are fine-tuned to further better their performance for specific tasks with supervised learning and RLHF – reinforcement learning from preference labeling feedback from humans (Stiennon et al., 2020; Ouyang et al., 2022). However, issues arise with RLHF where human feedback can become costly. To work around this problem, Reinforcement Learning from AI Feedback (RLAIF) was proposed (Bai et al., 2022b; Lee et al., 2023), which replaces human feedback with inexpensive feedback from other LLMs.

In many cases, the question lack a clear-cut “correct answer” and require creativity and imagination. As evident in an example of feedback by an LLM provided in Figure 1, when LLMs are tasked to assess responses to such prompts, the evaluation process can become arbitrary and introduce various biases. One prominent bias is the verbosity bias, which occurs when LLMs are influenced

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Judge Prompt

Please act as an impartial judge and evaluate the quality of the responses provided by two AI assistants to the user question displayed below. ... After providing your explanation, output your final verdict by strictly following this format: `[[A]]` if assistant A is better, `[[B]]` if assistant B is better, and `[[C]]` for a tie.

[User Question]

Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions.

[The Start of Assistant A's Answer]

I recently had the opportunity to travel to Hawaii, and it was an experience that I will never forget. From stunning beaches to vibrant culture,...

[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]

Hawaii is a tropical paradise, and I recently had the opportunity to experience it for myself. This archipelago is made up of eight main islands, each with its unique culture, traditions, and breathtaking landscapes ...

[The End of Assistant B's Answer]



Verdict

Both assistants did a great job in composing an engaging travel blog post about a recent trip to Hawaii. They both highlighted cultural experiences and must-see attractions, as requested by the user. ... While both responses are well-written and engaging, Assistant B's response is more detailed and provides a wider range of experiences, making it more engaging and informative for potential travelers. Therefore, Assistant B's response is better.

Final verdict: `[[B]]`

Figure 1: An example of a prompt to an LLM to judge two texts and the verdict. There is no one correct answer, and a comprehensive judgement is required.

by verbosity, favoring longer and more verbose texts, even if they appear wordy or of lower quality. Without accounting for this bias, LLM agents may learn to generate unnecessarily long texts. This may result in failures in downstream tasks such as lengthy summarizations or chatbots that return verbose responses to simple questions.

While previous studies have explored the concept of verbosity bias, they have tended to focus on specific cases. Zheng et al. (2023) limits their problem setting to questions answered with lists in their experiment on verbosity bias, and Huang et al. (2023) conducted experiments on summarization tasks. Moreover, these do not compare the preferences of LLMs to those of humans. We believe that such a comparison is crucial in challenging the conjecture that longer answers are inherently better and that LLMs are actually correct in their preferences.

Our contributions. In this paper, we conduct experiments on verbosity bias and saw that 1) LLMs exhibit a preference for longer answers in creative writing tasks, and 2) there is a discrepancy between of LLMs and those of humans in verbosity preference. Additionally, we formulate a quantification for measuring verbosity bias based on accuracy parity. This can be used to compare LLMs on their degree of verbosity bias.

2 Preliminaries

After undergoing pretraining for general purposes, LLMs are fine-tuned to further improve their performance in specific tasks. Pretraining is accomplished through self-supervised learning, where the model is trained to predict the next token in a sentence. Once the LLM is able to generate cohesive sentences, we proceed to fine-tune the model to solve specific tasks. One approach to fine-tuning involves supervised learning using expert data. This method relies on examples where experts have solved the task at hand. An example of a conversational LLM trained solely using this approach is Vicuna (Chiang et al., 2023). Vicuna achieved performance comparable to ChatGPT by utilizing user-shared conversations with ChatGPT as expert data. However, it is worth noting that obtaining expert data is often challenging.

RLHF addresses the challenge of limited training data in supervised learning by leveraging human feedback (Stiennon et al., 2020; Ouyang et al., 2022). This approach not only mitigates data scarcity but also significantly enhances alignment with human preferences, a critical factor in applications such as question answering. In RLHF, a reward model is trained to closely match human feedback data, which acts as the reward signal in the subsequent RL phase. Prominent LLMs like ChatGPT and Bard adopt a hybrid approach, combining both supervised learning and RLHF techniques to further refine their alignment with human preferences.

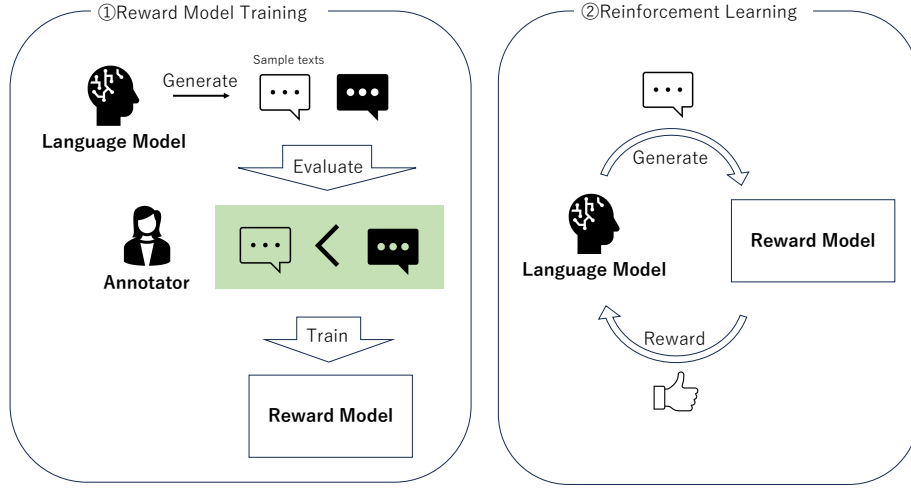


Figure 2: Two phases in RLHF. First the reward model is trained to align with human preference by matching with human feedback. In the second RL phase, the trained reward model provides reward signals to the language model.

2.1 RLHF

The first step of RLHF is to fit the reward function to align with human feedback. RL directly from human feedback as reward signal is unstable and requires volume. Therefore, a reward model that acts as the reward signal later in the process is trained to be consistent with human preference. Given a dataset $\mathcal{D}$ consisting of the original question, a pair of generated text, and the human preference label on which is chosen and rejected, the reward model is trained by minimizing

$$\mathcal{L}(\phi) = - \mathbb{E}_{(x, y_{\text{chosen}}, y_{\text{rejected}}) \sim \mathcal{D}} [\log \sigma(r_{\phi}(y_{\text{chosen}} | x) - r_{\phi}(y_{\text{rejected}} | x))], \quad (1)$$

where x is the prompt to the LLM, y_{chosen} is the preferred text, y_{rejected} is the rejected text, and r_{ϕ} parametrised with ϕ is the reward model that takes text as input and outputs the rating score.

In the second step of RLHF, now that we have a reward model to evaluate a generated text without human interaction, regular RL can take place. In this context, the state is the question and the generated text so far, action is the next token to generate, and the reward is $r_{\phi}(y)$ given after the full text is generated. This is equivalent to a task where a sparse reward is given only at episode termination. The LLM maximizes the signal from the reward model with the following:

$$\max_{\theta} \mathbb{E}[r_{\phi}(\pi_{\theta}(x) | x)], \quad (2)$$

where π_{θ} is a policy parameterized by θ . An optional KL divergence term is added to penalize the policy from deviating from the original policy.

2.2 RLAI

While RLHF brings down the cost of human labor compared to generating an expert data from scratch, human feedback is still costly. For example, in Wang et al. (2023), it cost around 3 minutes (\$0.75 if \$15.00 per hour) per evaluation. In one occasion, OpenAI worked around this by employing people in Kenya on less than \$2 per hour pay in the process of labeling violent or inappropriate texts. They were under scrutiny for the unideal working conditions.

To combat these problems, RLAI was proposed. This method replaces human feedback with feedback from other LLMs. This brings down the cost significantly; in our case, evaluation cost around \$0.05 each, which is 1/15th compared to human feedback in the previously cited paper (Wang et al., 2023).

2.3 Biases in Automated LLM Evaluation

When LLMs evaluate generated texts, various biases are introduced. We provide below a list of biases discussed in various papers (Zheng et al., 2023; Bai et al., 2022a; Wang et al., 2023).

Position Bias: Position bias occurs when, in comparing generated texts, LLMs prefer the answer given in certain positions. If we define the ground truth probability of a (the first parameter) preferred over b (the second parameter) to be $P(a, b)$, it should be that $P(y_0, y_1) = 1 - P(y_1, y_0)$, meaning the position should have no affect on the judgement. Position bias is when the comparison by model is $\hat{P}(y_0, y_1) \neq (1 - \hat{P}(y_1, y_0))$. For example, GPT-4 tends to prefer the first option given to it, while ChatGPT prefers the second option (Wang et al., 2023). To account for this bias, we can simply swap the positions and evaluate the options twice. If the model gives contradicting results between permutations, we count it as a draw.

Wang et al. (2023) has proposed several methods to calibrate this bias further: Multiple Evidence Calibration asks the LLM to provide evidence before making judgement, and Human-in-the-Loop Calibration involves human adjustment when deemed necessary.

Self-enhancement Bias: LLMs tend to prefer answers generated by itself compared to answers generated by other models. This becomes a problem when benchmarking LLMs by evaluating them with LLMs (Zheng et al., 2023), but not so much in the context of RLAIIF, as the comparisons are always between answers generated by the same model.

Verbosity Bias: Verbosity bias refers to the bias where LLMs prefer longer, more verbose answers even if there are no difference in quality. Training with RLAIIF with verbosity bias present can lead to LLMs generating excessively long responses, when in reality a much more concise response would suffice. In tasks such as question answering, a verbose response can be critical to its usefulness, but there aren't enough researches that look into this. For these reasons we take a closer look into this.

There are several proposed methods to mitigate the effect of biases.

Chain-of-thought Prompting is a prompting technique where the LLM is asked to provide the thought process before generating the actual evaluation. This way, at the time when the LLM generates the actual evaluation, it has its chain-of-thought to base its evaluation from. This encourages human alignment and more accurate evaluations, rather than arbitrary evaluations without thought.

One-shot/Few-shot Prompting is another prompting technique which gives one example/several examples of a prompt and its corresponding correct answer when prompting the LLM. When generating the response, the LLM can continue the pattern from the examples to better align with the intended response.

3 Related Works

3.1 RLAIIF Advancements

There have been several recent advancements in the field of RLAIIF. Bai et al. (2022b) trained an LLM via RLAIIF with limited human feedback. In this work, they claim that helpfulness and harmfulness have a trade-off relationship, and aim to train an LLM that keeps a balance between those two. Their method only requires human feedback in the helpfulness aspect, and harmless behavior is achieved purely from RLAIIF. The LLMs trained in the work by Lee et al. (2023) achieved near-human performance in summarization tasks with RLAIIF without any human feedback. While not a study on RLAIIF itself, Zheng et al. (2023) evaluate LLMs with other LLMs as a judge and show that GPT-4 has a high human alignment and agrees with humans on over 80% of evaluations.

3.2 On Verbosity Bias in Evaluations by LLMs

Zheng et al. (2023) also provides lists of biases and methods to overcome them. Alongside their experiment on position bias, they experimented on verbosity bias by attempting a "repetitive list attack" on several LLMs. This attack pertains to "listing" tasks, in which the prompt asks to list several items (e.g. "What are examples of fruits that are round?"). The "repetitive list attack" is done by making the answers verbose by repeating items multiple times, and then asking the LLMs to evaluate these augmented answers. If the LLM evaluates these "repetitive lists" to be better than the

original, the attack is considered a success. Their results show GPT-4 is significantly less prone to this attack with below 10% success rate, while GPT-3.5 and Claude-v1 both suffer over 90% success rate. Compared to this research, we expand the problem setting to general question-answering tasks. Huang et al. (2023) tackle verbosity bias in summarization tasks. They found that GPT-4 actually prefers short responses in faithfulness and coverage when it comes to summarization, although this is seen strongly only in single-answer grading, and not in comparison grading. This suggests that verbosity bias can be different between different tasks.

Compared to these studies, our problem setting is more general and we compare the verbosity preference between humans and LLMS. The experiments conducted in these papers measure the difference in evaluations when the texts are artificially made verbose while maintaining the same content. The assumption is that elongating the texts would have no effect on a true evaluator, so the difference in evaluation indicates verbosity bias. In our attempt to broaden the problem setting, we make use of human feedback as the oracle instead of making this assumption.

4 Verbosity Preference of LLMS

First, we experiment to see how much LLMS actually prefer longer answers. We ask GPT-4 to choose between pairs of responses and examine if it prefers longer responses or not. We did not limit our scope to prompts answered with a specific format (like lists in Zheng et al. (2023)) in order to observe the LLMS’ general tendency to prefer longer answers.

We generated 100 sample answers each to 3 prompts, all from the same model (Vicuna-7b-v1.5) generated with the temperature parameter set to 0.7. One of the questions and two examples of the answers are as follows.

- Question: *Compose an engaging travel blog post about a recent trip to Hawaii, highlighting cultural experiences and must-see attractions*
 1. I recently had the opportunity to travel to Hawaii, and it was an experience that I will never forget. From stunning beaches to vibrant culture, there was so much to explore and enjoy during my time on the islands...
 2. Hawaii is a tropical paradise, and I recently had the opportunity to experience it for myself. This archipelago is made up of eight main islands, each with its unique culture, traditions, and breathtaking landscapes. During my trip, I had the chance to visit several cultural sites, such as the Polynesian Cultural Center on Oahu...
 3. ...

The prompts are taken from the library introduced by Zheng et al. (2023), all from the "creative" category because 1) answers generated to other categories didn’t vary in word count enough to see verbosity bias and 2) GPT-4 was not good at judging answers in those categories. We then take answers from these generated samples and insert them into the template shown in Figure 1. With the template complete, we asked GPT-4 to evaluate preferences between pairs of answers with the template. The outcome is either the first option selected, a draw, or the second option selected. In order to account for position bias, GPT-4 evaluated the pair twice with the position swapped the second time. It was considered a draw unless it gave the same result on both permutations.

The results are shown in Figure 3a for the overall result, and Figures 3b to 3d for results from each prompt. Both in the overall result and the individual results, there is a tendency for GPT-4 to prefer longer answers. When the word count difference is large enough, GPT-4 almost always prefers the longer answer. For question 1 and 2, the preference is smooth and clear, while for question 3, when the word count difference is small, there is high variance in evaluation. As we can see the shape varies between questions, and therefore we can deduce that verbosity does not rely entirely on word count and is different for each question. This makes adjusting for verbosity post-evaluation hard unless we know the verbosity preference shape for the prompt in question.

From this experiment, we can draw the conclusion that GPT-4 generally prefers longer answers among those that are generated by the same LLM with the same prompt. However, this experiment by itself does not indicate that GPT-4 suffers from verbosity bias; it could be that the longer answers generated by vicuna are actually higher in quality and helpfulness. In order to truly measure verbosity bias, we would need the ground truth of each comparison which we do not have. Instead, we next utilize a dataset of human evaluations as the baseline.